

EXECUTIVE SUMMARY

Data Engineer and AI/ML Engineer with over seven years of expertise building robust data pipelines and production machine learning and generative AI systems. Optimized distributed systems, ensuring scalable infrastructure deployment with hands-on experience deploying ML models and LLM/RAG-based AI applications into production using Spark, Airflow, Kubernetes, Docker, and AWS/Azure cloud platforms. Skilled in designing scalable backend systems, CI/CD automation, containerization, MLOps, and time-series forecasting solutions for large-scale analytics engineering and optimization workloads. Experienced in PySpark, TensorFlow, PyTorch, Hadoop, SQL, Generative AI, and cloud-native data engineering and AI engineering best practices.

Core Skills: Data Engineering, AI Engineering, Machine Learning Engineering, Analytics Engineering, ETL/ELT Pipeline Design, Data Pipeline Architecture, MLOps, LLM & RAG Systems, Generative AI, Spark, PySpark, Airflow, Kubernetes, Docker, AWS, Azure, GCP, Snowflake, Redshift, Databricks, TensorFlow, PyTorch, SQL, Python, CI/CD Automation.

PROFESSIONAL EXPERIENCE

Principal Data Engineer | ESTUATE

April 2024 - Present

Client: PRIMIENT

- Led and mentored a cross-functional team of 10+ data engineers, analysts, and developers in delivering scalable enterprise analytics and data platform solutions.
- Designed USA Federal Database for Economic Trends and Analysis from multiple Government Analyses for Upcoming 5 years' production planning.
 - Delivered predictive analytics and operational dashboards for engineering and business teams.
- Reduced compute processing, saving costs by 10% via efficient data staging, Transformation in data solutions, and optimization techniques.
- Performed advanced root cause analysis for data quality, extraction, transformation, and integration issues across SAP, Workday, SharePoint, and third-party systems, significantly reducing recurring production incidents.
- Built secure API integrations between Azure AI Foundry, enterprise applications, and data platforms to support AI-powered workflows and automation initiatives.
- Designed and optimized enterprise-scale data pipelines supporting manufacturing, supply chain, inventory, and commodity analytics for large-scale industrial operations.
- Championed Agile and Scrum methodologies, driving sprint planning, backlog prioritization, and continuous improvement initiatives across data engineering teams.
- Established monitoring, alerting, and operational excellence practices that improved platform uptime and accelerated incident resolution.
- Developed Retrieval-Augmented Generation (RAG) architectures utilizing Azure AI Foundry, Azure OpenAI, vector databases, and enterprise knowledge repositories as part of AI engineering initiatives to improve AI response accuracy and contextual relevance.

Client: Enphase Energy

- Developed and enhanced API-driven integrations for near real-time data ingestion, reducing manual intervention and improving data availability for business users.
- Built scalable backend systems for processing high-volume IoT and inverter data streams, enabling real-time performance monitoring and predictive maintenance analytics
- Designed complex SQL queries and data infrastructure within Incoarta for critical reporting on Revenue, Sales, and Inventory.
- Optimized Redshift and Snowflake data warehouse environments through partitioning, clustering, materialized views, and workload management, significantly improving query performance.
- Implemented data orchestration workflows using Apache Airflow to manage complex dependencies across hundreds of data ingestion, transformation, and machine learning pipelines.

- Built RESTful APIs and event-driven integrations supporting secure exchange of operational, customer, and energy telemetry data between enterprise systems and cloud platforms.
- Developed feature engineering pipelines

Client: Zuora

- Developed time-series forecasting and optimization models for production planning, commodity pricing, demand forecasting, and operational efficiency improvements.
- Designed data pipelines using AWS Glue, Lambda, Redshift, Athena, and S3, enabling scalable processing of medical equipment sales, inventory, warranty, and service datasets.
- Developed and maintained automated workflows in Dataiku DSS to support data ingestion, transformation, validation, and delivery across business-critical reporting systems.
- Collaborated with data scientists and business analysts to support AI engineering and machine learning engineering initiatives using AWS SageMaker, Amazon Bedrock, and Azure AI Foundry.
- Built AI-powered analytics solutions integrating enterprise sales, inventory, and service data to support demand forecasting, customer segmentation, and predictive maintenance use cases.
- Developed secure REST APIs and event-driven integrations using AWS services and enterprise applications to support real-time data exchange and workflow automation.
- Implemented scalable ETL/ELT pipelines using Python, SQL, Spark, Dataiku, AWS Glue, and Lambda, significantly improving data processing efficiency and reliability.

Data Engineer | Lowe's

April 2023 - March 2024

- Architected and implemented a distributed data pipeline for a new product launch, leveraging Test-Driven Development to ensure scalability and performance.
- Managed and documented all data sources, including Oracle ERP and CRM, enabling efficient data ingestion for product innovation and performing rigorous data quality testing.
- Cleansed and transformed diverse data formats using Databricks, improving data quality for downstream analytics and ensuring adherence to data management best practices.
- Achieved 20% faster data processing via optimized distributed system design improved data reliability and reduced testing time through efficient data management practices.

Data Engineer | Pfizer

May 2022 - March 2023

- Created data pipelines with Airflow and Spark for critical data analysis, deploying solutions on Google Cloud Platform and ensuring robust Data Ops.
- Built a comprehensive data warehouse with SQL Server for historical data analysis, effectively utilizing Cloud Storage and managing diverse Data Storage needs.
- Developed Snowflake solutions and stream processing jobs with Spark utilized Docker for containerization, improving deployment speed and reliability.
- Generated insightful data visualizations using Tableau and Looker, translating complex data into actionable business intelligence.

Data Engineer | CBRE

April 2019 - December 2020

- Developed and deployed a scalable backend service for data management, processing terabytes of data daily with high availability.
- Designed and implemented ETL pipelines using Spark and Hadoop to prepare data for analysis, leveraging Go for backend services and ensuring data quality
- Developed a new data cleansing algorithm, reducing error rates by 15%, enhancing Interactive Media platform efficiency and demonstrating Data Engineering expertise.
- Ensured compliance with enterprise data governance policies using Azure Purview for data cataloging and lineage tracking.
- Built real-time data ingestion pipelines using Azure Event Hubs for handling streaming data from multiple sources.

- Developed and deployed a scalable backend service for data management, processing terabytes of data daily with high availability.
- Designed and implemented ETL pipelines using Spark and Hadoop to prepare data for analysis, leveraging Go for backend services and ensuring data quality.
- Developed a new data cleansing algorithm, reducing error rates by 15%, enhancing Interactive Media platform efficiency and demonstrating Data Engineering expertise.
- Ensured compliance with enterprise data governance policies using Azure Purview for data cataloging and lineage tracking.

Process Developer | Genpact

September 2017 - March 2019

- Designed and implemented ETL pipelines using Apache Airflow for Interactive Media content, supporting Data Engineering team with reliable data flows.
- Optimized PySpark DataFrame operations within a Hadoop ecosystem, increasing Interactive Media data ingestion speed by 20%.
- Optimized PySpark DataFrame operations within a Hadoop ecosystem, increasing Interactive Media data ingestion speed by 20%.
- Transformed raw data into usable formats for Media Foundation analytics using Spark and Hadoop, improving data quality for Computer Information System applications.
- Transformed raw data into usable formats for Media Foundation analytics using Spark and Hadoop, improving data quality for Computer Information System applications.

Certifications

- | | |
|--|---|
| • Data Bricks Generative AI Engineer Associate | • Google Cloud Big Query & Looker Studio |
| • Apache Iceberg | • Oracle Cloud Infrastructure Foundations Associate |
| • Incorta Developer | • Microsoft Azure Data Engineer Associate |
| • Dataiku Core Designer | • Certified Pega System Architect |
| • Dremio | • AWS Cloud Certification |